

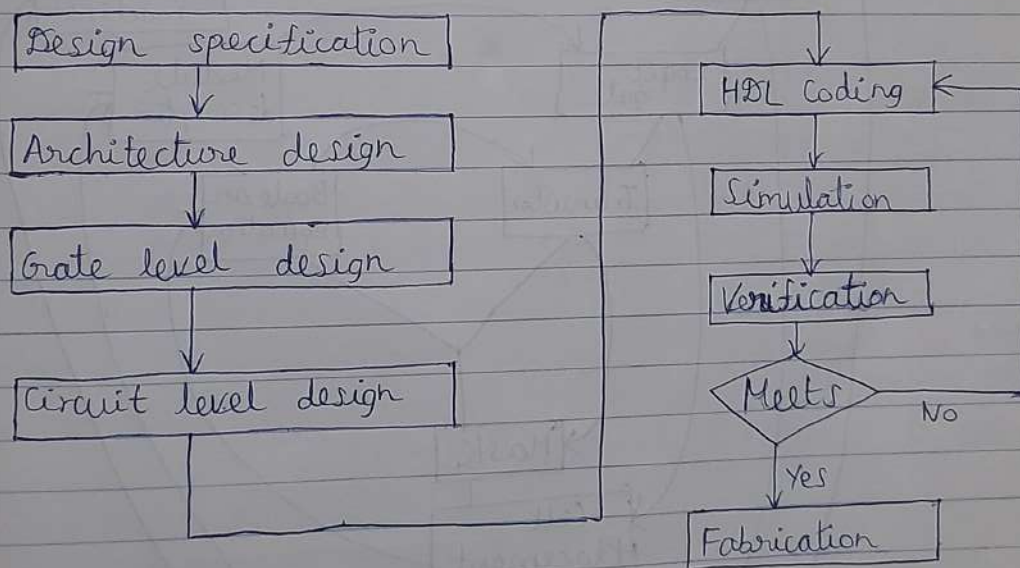
VLSI Design Flow:

Basics of VLSI design flow:

> The design flow start with a given set of specification or requirements.

> At the end, when the desired specifications are not met, the design has to be modified & re-checked several times until it meets the required specifications.

Flow chart of VLSI design flow:

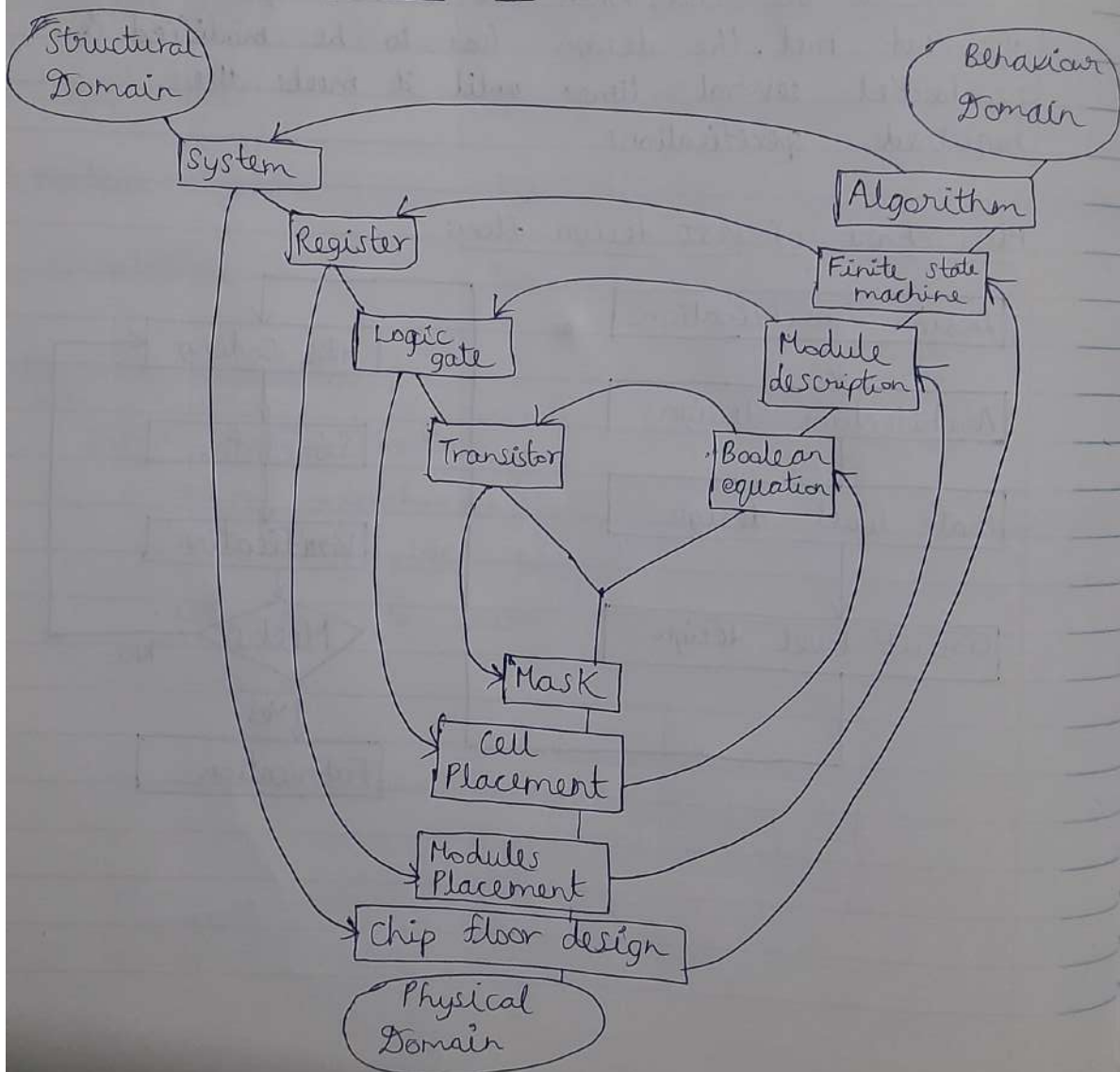


Domains of VLSI design flow

The design description for VLSI circuit be described in terms of 3 domains:

- > Behavioral domain
- > Structural domain
- > Physical domain

Y chart of VLSI design



Digital logic fundamentals

- > Logic gates are the fundamental building blocks of digital electronics.
- > They take one or more binary inputs (0 and 1) to produce single binary output.

Basic logic gates

- AND Gate $\rightarrow A \cdot B = Y$
Output 1 if both inputs are 1.
- OR Gate $\rightarrow A + B = Y$
Output 1 if ^{at least} one input is 1.
- NOT Gate $\rightarrow Y = \bar{A}$
It takes single input and reverses it.

Universal gate

- NAND Gate $\rightarrow Y = \overline{A \cdot B}$
This is an AND gate followed by NOT gate.
- NOR Gate $\rightarrow Y = \overline{A + B}$
This is an OR gate followed by NOT gate.

The special gate

- XOR gate $\rightarrow Y = A \oplus B$
Output 1 if the inputs are different.

• XNOR Gate $\rightarrow Y = \overline{A \oplus B}$

* Output 1 when inputs are same

* Inverse of XOR gate

Truth Table

1. AND Gate $\Rightarrow Y = A \cdot B$

A	B	output (Y)
0	0	0
0	1	0
1	0	0
1	1	1

2. OR Gate $\Rightarrow Y = A + B$

A	B	output (Y)
0	0	0
0	1	1
1	0	1
1	1	1

3. NOT Gate $\Rightarrow Y = \bar{A}$

A	B	Output (Y)

Input (A)	Output (Y)
0	1
1	0

4. NAND Gate $\Rightarrow Y = \overline{A \cdot B}$

A	B	Output (Y)
0	0	1
0	1	1
1	0	1
1	1	0

5. NOR Gate $\Rightarrow Y = \overline{A + B}$

A	B	Output (Y)
0	0	1
0	1	0
1	0	0
1	1	0

6. XOR Gate $\Rightarrow Y = A \oplus B$

A	B	Output (Y)
0	0	0
0	1	1
1	0	1
1	1	0

7. XNOR Gate $\Rightarrow Y = \overline{A \oplus B}$

A	B	Output (Y)
0	0	1
0	1	0
1	0	0
1	1	1

Observations from simulations

Simulate AND gate $\Rightarrow$

By simulating this gate, I observed by changing the binary inputs (1 and 0). If both inputs are 1, output is 1. If one input is 0 and other input is 1, output is 0.

Simulate OR gate $\Rightarrow$

If one input is 1 and other input is 0, both the inputs are 1 gives the output as 1.

If both inputs are as 0 it gives the output as 0.

Simulate NOT gate $\Rightarrow$

The output is inverse of the given input.

Eg: If input is 1, output is 0

input is 0, output is 1

simulate NAND gate $\Rightarrow$

By simulating NAND gate, it behaves as the exact opposite of AND gate.

simulate NOR gate $\Rightarrow$

By simulating, it behaves as the exact opposite of OR gate.

simulate XOR gate $\Rightarrow$

By simulating XOR gate, If the both inputs are same it gives the output as 0. If the two inputs are different it gives the output as 1.

simulate XNOR gate $\Rightarrow$

It behaves as the exact opposite of XOR gate.

simulate Half adder circuit $\Rightarrow$

By simulating this circuit, it has sum and carry as outputs. If two inputs are given as 1 it gives the output of sum as 0 and output of carry as 1.